

Anesthesia / OT Technician MCQs

Part 10 - Preparation for AIIMS CRE, GIMS, NCL, BTSC Exams

1. Which device helps remove waste anesthetic gases from OT?

- A. Humidifier
- B. Scavenging system
- C. Flow meter
- D. Vaporizer

Answer: B. Scavenging system

Explanation: The scavenging system is used to remove waste anesthetic gases from the operating theater to protect personnel from exposure.

2. Which anesthetic machine component should be checked daily for leaks?

- A. OT table
- B. Defibrillator
- C. Suction apparatus
- D. Vaporizer and breathing circuit

Answer: D. Vaporizer and breathing circuit

Explanation: The breathing circuit and vaporizer are the most prominent sites where gas leaks can occur, so they must be checked daily.

3. Which component delivers oxygen rapidly during an emergency?

- A. Oxygen flush valve
- B. Flow meter
- C. Humidifier
- D. Pressure gauge

Answer: A. Oxygen flush valve

Explanation: The oxygen flush valve (also known as bypass valve) delivers a large amount of oxygen (35-75 L/min) directly to the breathing circuit, bypassing the vaporizer, in emergencies like severe hypoxia.

4. During anesthesia, a sudden absence of EtCO₂ waveform after intubation most commonly indicates:

- A. Hyperthermia
- B. Pulmonary edema
- C. Hypertension
- D. Esophageal intubation

Answer: D. Esophageal intubation

Explanation: A zero EtCO₂ waveform after intubation means no carbon dioxide is being exhaled, which indicates the tube was mistakenly placed in the esophagus instead of the trachea.

5. Which part of the anesthesia machine is located between flow meters and common gas outlet?

- A. Pressure regulator
- B. Vaporizer
- C. Oxygen cylinder
- D. Scavenging system

Answer: B. Vaporizer

Explanation: The typical sequence in an anesthesia machine is Flow meter -> Vaporizer -> Common gas outlet. The vaporizer converts liquid anesthetic into gas.

6. Which complication may occur due to excessive airway pressure during ventilation?

- A. Barotrauma
- B. Hypoglycemia
- C. Bradycardia
- D. Hyperkalemia

Answer: A. Barotrauma

Explanation: Excessive airway pressure can cause the alveoli in the lungs to over-expand and rupture, a condition known as barotrauma.

7. Which monitor gives important information about patient ventilation and metabolism together?

- A. ECG
- B. Pulse oximeter
- C. Capnograph
- D. Thermometer

Answer: C. Capnograph

Explanation: Capnography measures EtCO₂, indicating both ventilation (exhalation of CO₂) and cellular metabolism (production of CO₂).

8. Assertion (A): Oxygen flush bypasses vaporizer. Reason (R): Oxygen flush delivers oxygen directly to breathing circuit at high flow.

- A. Both A and R are true and R explains A
- B. Both A and R are true but R does not explain A
- C. A is true but R is false
- D. A is false but R is true

Answer: A. Both A and R are true and R explains A

Explanation: The oxygen flush valve bypasses the vaporizer to deliver a high flow of pure oxygen directly to the patient's breathing circuit during emergencies.

9. Which instrument would be most inappropriate for robotic surgery setup?

- A. Endowrist instrument
- B. Trocar cannula
- C. Hudson brace
- D. Robotic bipolar grasper

Answer: C. Hudson brace

Explanation: A Hudson brace is an orthopedic/neurosurgical instrument used for making burr holes in the skull, making it entirely inappropriate for a robotic surgery setup, whereas the other three are standard robotic laparoscopic tools.

10. Which cavity contains lungs?

- A. Abdominal cavity
- B. Pelvic cavity
- C. Thoracic cavity
- D. Cranial cavity

Answer: C. Thoracic cavity

Explanation: The thoracic cavity contains the lungs and heart. The space between the lungs where the heart is located is the mediastinum.

11. Which side of the heart pumps blood to the lungs?

- A. Left side
- B. None
- C. Both sides
- D. Right side

Answer: D. Right side

Explanation: The right side (right ventricle) pumps deoxygenated blood to the lungs via the pulmonary artery for oxygenation.

12. Which structure covers the ends of bone in movable joints?

- A. Fibrous tissue
- B. Hyaline cartilage
- C. Tendon
- D. Fascia

Answer: B. Hyaline cartilage

Explanation: Hyaline cartilage is a flexible structure that covers the ends of bones in movable joints, helping in smooth movement.

13. Which organ stores bile?

- A. Liver
- B. Pancreas
- C. Gall bladder
- D. Spleen

Answer: C. Gall bladder

Explanation: While the liver produces bile, it is stored and concentrated in the gall bladder.

14. Which blood group is called universal donor?

- A. AB positive
- B. O positive
- C. O negative
- D. AB negative

Answer: C. O negative

Explanation: O negative blood has no A, B, or Rh antigens, making it the universal donor that can be given to any blood type.

15. Which organ produces insulin?

- A. Pancreas
- B. Gall bladder
- C. Liver
- D. Kidney

Answer: A. Pancreas

Explanation: The pancreas produces insulin, specifically from the beta cells in the islets of Langerhans, to regulate blood glucose levels.

16. Which joint is ball and socket joint?

- A. Elbow joint
- B. Knee joint
- C. Shoulder joint
- D. Interphalangeal joint

Answer: C. Shoulder joint

Explanation: The shoulder joint (along with the hip joint) is a ball and socket joint, allowing for a wide range of motion.

17. Which structure carries urine from kidney to bladder?

- A. Urethra
- B. Ureter
- C. Vas deferens
- D. Renal vein

Answer: B. Ureter

Explanation: The ureters are duct-like structures that transport urine from the hilum of the kidneys to the urinary bladder.

18. Which bone forms the shin of the leg?

- A. Fibula
- B. Femur
- C. Tibia
- D. Patella

Answer: C. Tibia

Explanation: The tibia is the larger, weight-bearing bone in the lower leg, commonly known as the shin bone.

19. Which ethical principle means 'do no harm'?

- A. Justice
- B. Fidelity
- C. Beneficence
- D. Non-maleficence

Answer: D. Non-maleficence

Explanation: Non-maleficence is the ethical principle of 'primum non nocere', which means first, do no harm to the patient.

20. Which device is used for manual ventilation during respiratory arrest?

- A. Ambu bag
- B. Nebulizer
- C. Defibrillator
- D. Infusion pump

Answer: A. Ambu bag

Explanation: An Ambu bag (Artificial Manual Breathing Unit) is a hand-held device used to provide positive pressure ventilation to patients who are not breathing or not breathing adequately.

21. Which symptom is commonly seen in pneumothorax?

- A. Severe breathlessness
- B. Hematemesis
- C. Polyuria
- D. Pedal edema

Answer: A. Severe breathlessness

Explanation: Pneumothorax restricts lung inflation, leading to difficulty in breathing, which presents as severe breathlessness.

22. Which machine removes secretions from airway?

- A. Ventilator
- B. Defibrillator
- C. Suction machine
- D. Infusion pump

Answer: C. Suction machine

Explanation: A suction machine is used to clear the airway by removing saliva, blood, or other secretions.

23. Maximum Glasgow Coma Scale (GCS) score?

- A. 10
- B. 12
- C. 15
- D. 20

Answer: C. 15

Explanation: The maximum GCS score is 15 (Eye 4, Verbal 5, Motor 6), indicating a fully alert and oriented patient. The minimum is 3.

24. Bed sores commonly develop over?

- A. Fatty tissue
- B. Bony prominences
- C. Hair bearing area
- D. Finger tips

Answer: B. Bony prominences

Explanation: Bony prominences compress blood vessels against the bed surface, decreasing blood supply and causing tissue death, leading to bed sores (pressure ulcers).

25. Universal precautions are mainly used to prevent?

- A. Fracture
- B. Diabetes
- C. Hypertension
- D. Hospital acquired infection

Answer: D. Hospital acquired infection

Explanation: Universal precautions maintain hygiene to prevent the spread of nosocomial (hospital-acquired) infections.

26. What does PPE stand for?

- A. Personal Protective Equipment
- B. Patient Protective Equipment
- C. Professional Prevention Equipment
- D. Personal Patient Equipment

Answer: A. Personal Protective Equipment

Explanation: PPE stands for Personal Protective Equipment, which includes gear like masks, caps, and gloves to protect healthcare workers.

27. Which airway device enters directly into trachea?

- A. LMA (Laryngeal Mask Airway)
- B. Nasal cannula
- C. ET tube (Endotracheal tube)
- D. Face mask

Answer: C. ET tube (Endotracheal tube)

Explanation: An endotracheal (ET) tube is directly inserted into the trachea to maintain an open airway, whereas others sit externally or supraglottically.